

2c. Content of Algorithms and programming (Component 02)

This component will incorporate and build on the knowledge and understanding gained in the Computer systems component (01).

In addition, learners should:

- understand what is meant by computational thinking
- understand the benefits of applying computational thinking to solving a wide variety of problems
- understand the principles of solving problems by computational methods
- be able to use algorithms to describe problems
- be able to analyse a problem by identifying its component parts.

2.1 Elements of computational thinking	
Understand what is meant by computational thinking	
2.1.1 Thinking abstractly	(a) The nature of abstraction. (b) The need for abstraction. (c) The differences between an abstraction and reality. (d) Devise an abstract model for a variety of situations.
2.1.2 Thinking ahead	(a) Identify the inputs and outputs for a given situation. (b) Determine the preconditions for devising a solution to a problem. (c) The nature, benefits and drawbacks of caching. (d) The need for reusable program components.
2.1.3 Thinking procedurally	(a) Identify the components of a problem. (b) Identify the components of a solution to a problem. (c) Determine the order of the steps needed to solve a problem. (d) Identify sub-procedures necessary to solve a problem.
2.1.4 Thinking logically	(a) Identify the points in a solution where a decision has to be taken. (b) Determine the logical conditions that affect the outcome of a decision. (c) Determine how decisions affect flow through a program.
2.1.5 Thinking concurrently	(a) Determine the parts of a problem that can be tackled at the same time. (b) Outline the benefits and trade offs that might result from concurrent processing in a particular situation.